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$$f_1(x, y, z) = (xy)^z$$

$$f_2(x, y, z) = x^{yz}$$

$$f_3(x, y, z) = x^{y^z}$$

$$\frac{\partial f_1}{\partial x} = z(xy)^{z-1}y, \quad \frac{\partial f_1}{\partial y} = z(xy)^{z-1}x, \quad \frac{\partial f_1}{\partial z} = (xy)^z \ln |xy|$$

$$\frac{\partial f_2}{\partial x} = yz x^{yz-1}, \quad \frac{\partial f_2}{\partial y} = x^{yz} z \ln |x|, \quad \frac{\partial f_2}{\partial z} = x^{yz} y \ln x$$

$$\frac{\partial f_3}{\partial x} = y^z x^{y^z-1}, \quad \frac{\partial f_3}{\partial y} = x^{y^z} \ln |x| z y^{z-1}, \quad \frac{\partial f_3}{\partial z} = x^{y^z} \ln |x| y^z \ln |y|$$

$$\begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} & \frac{\partial f_1}{\partial z} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} & \frac{\partial f_2}{\partial z} \\ \frac{\partial f_3}{\partial x} & \frac{\partial f_3}{\partial y} & \frac{\partial f_3}{\partial z} \end{pmatrix} = \begin{pmatrix} z(xy)^{z-1}y & z(xy)^{z-1}x & (xy)^z \ln |xy| \\ yz x^{yz-1} & x^{yz} z \ln |x| & x^{yz} y \ln x \\ y^z x^{y^z-1} & x^{y^z} \ln |x| z y^{z-1} & x^{y^z} \ln |x| y^z \ln |y| \end{pmatrix}$$

References